Centrality in a Network

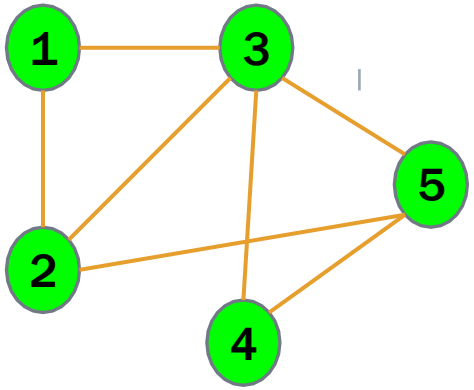
- Influential players often play central roles in a network
- Defining/Identifying influential players always remain hard
  - Some players attract limelight
  - Some others play behind the scene
  - Many others do important linkage
  - and so on…
- To identify influential players, we require
  - to define a notion of influence
  - to device measure that can capture that influence

Degree Centrality

- Centrality of the simplest kind
- In a sense, captures the popularity of a player within a network
- Quantifies the direct influence of a node on its local neighbourhood
- The degree centrality $C_d(v)$ of a node v in a network $G(V, E)$ is defined as:

$$C_d(v) = \frac{\deg(v)}{\max_{u \in V} \deg(u)}$$

- Particularly useful for marketing scenarios, wherein the detected influential user can promote a product/service across her followers
- Degree centrality of the nodes 1 through 5 in network G_1 are $^2/_4$, $^3/_4$, $^4/_4$, $^2/_4$, and $^3/_4$, respectively; i.e., 0.5, 0.75, 1.0, 0.5, and 0.75, respectively. So, node 3 is most central according to degree centrality measure.



G_1

Closeness Centrality

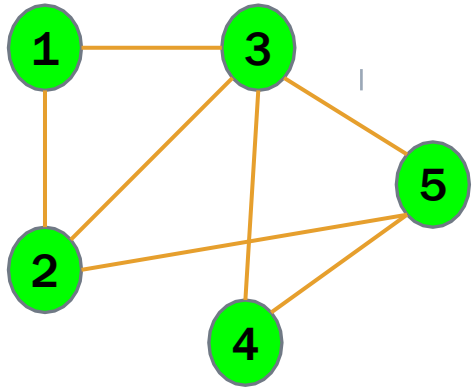
- A means for detecting nodes that can spread information very efficiently through a graph
- The measure is useful in
  - Examining/restricting the spread of fake news/misinformation in social media
  - Examining/restricting the spread of a disease in epidemic modelling
  - Controlling/restricting the flow of vital information and resources within an organization (a terrorist network, for example)
- The closeness centrality $C(v)$ of a node v in a network $G(V, E)$ is defined as

$$C(v) = \frac{|V| - 1}{\sum_{u \in V,\, u \neq v} d(u, v)}$$

Where $d(u, v)$ denotes the distance of node u from node v

- The measure indicates how close a node from the rest of the network

Closeness Centrality



G_1

- In graph G_1, the closeness centrality for the nodes are as follows

$$C(1) = \frac{5-1}{1+1+2+2} = \frac{4}{6} = 0.67$$

$$C(2) = \frac{5-1}{1+1+2+1} = \frac{4}{5} = 0.80$$

$$C(3) = \frac{5-1}{1+1+1+1} = \frac{4}{4} = 1.0$$

$$C(4) = \frac{5-1}{2+2+1+1} = \frac{4}{6} = 0.67$$

$$C(5) = \frac{5-1}{2+1+1+1} = \frac{4}{5} = 0.80$$

- Clearly, node 3 is most central according to closeness centrality measure

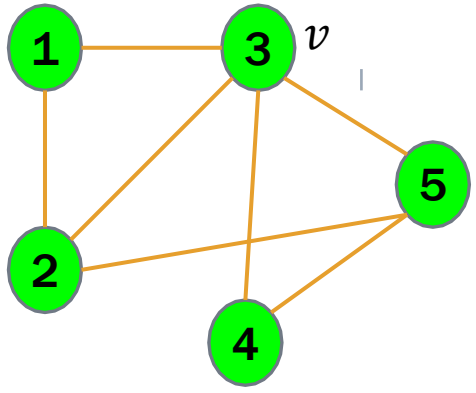
Betweenness Centrality

- A measure to compute how central a node is in between paths of the network
- A measure to compute how many (shortest) paths of the network pass through the node
- Useful in identifying
  - the articulation points, i.e., the points in a network which, if removed, may disconnect the network
  - The super spreaders in analyzing disease spreading in epidemiology
  - the suspected spies in security networks
- The betweenness centrality $C_B(v)$ of a node v in a network $G(V, E)$ is defined as

$$C_B(v) = \sum_{x,y \in V,\ x \neq v \neq y} \frac{\sigma_{xy}(v)}{\sigma_{xy}}$$

where σ_{xy} denotes the number of shortest paths between nodes x and y in the network, $\sigma_{xy}(v)$ denotes the same passing though v. If $x = y$, then $\sigma_{xy} = 1$.

Betweenness Centrality



G_1

- To find the betweenness centrality of node $v = 3$ in graph G_1
- The following matrix is of the form $\sigma_{xy}(v)|\sigma_{xy}$

$\sigma_{xy}(v)\|\sigma_{xy}$	1	2	3	4	5
1	0	0|1	–	1|1	1|2
2	0|1	0	–	1|2	0|1
3	–	–	–	–	–
4	1|1	1|2	–	0	0|1
5	1|2	0|1	–	0|1	0

- Thus the betweenness centrality of node 3 $= \frac{1}{1} + \frac{1}{2} + \frac{1}{2} + \frac{1}{1} + \frac{1}{2} + \frac{1}{2} = 4$

Betweenness Centrality: Variants

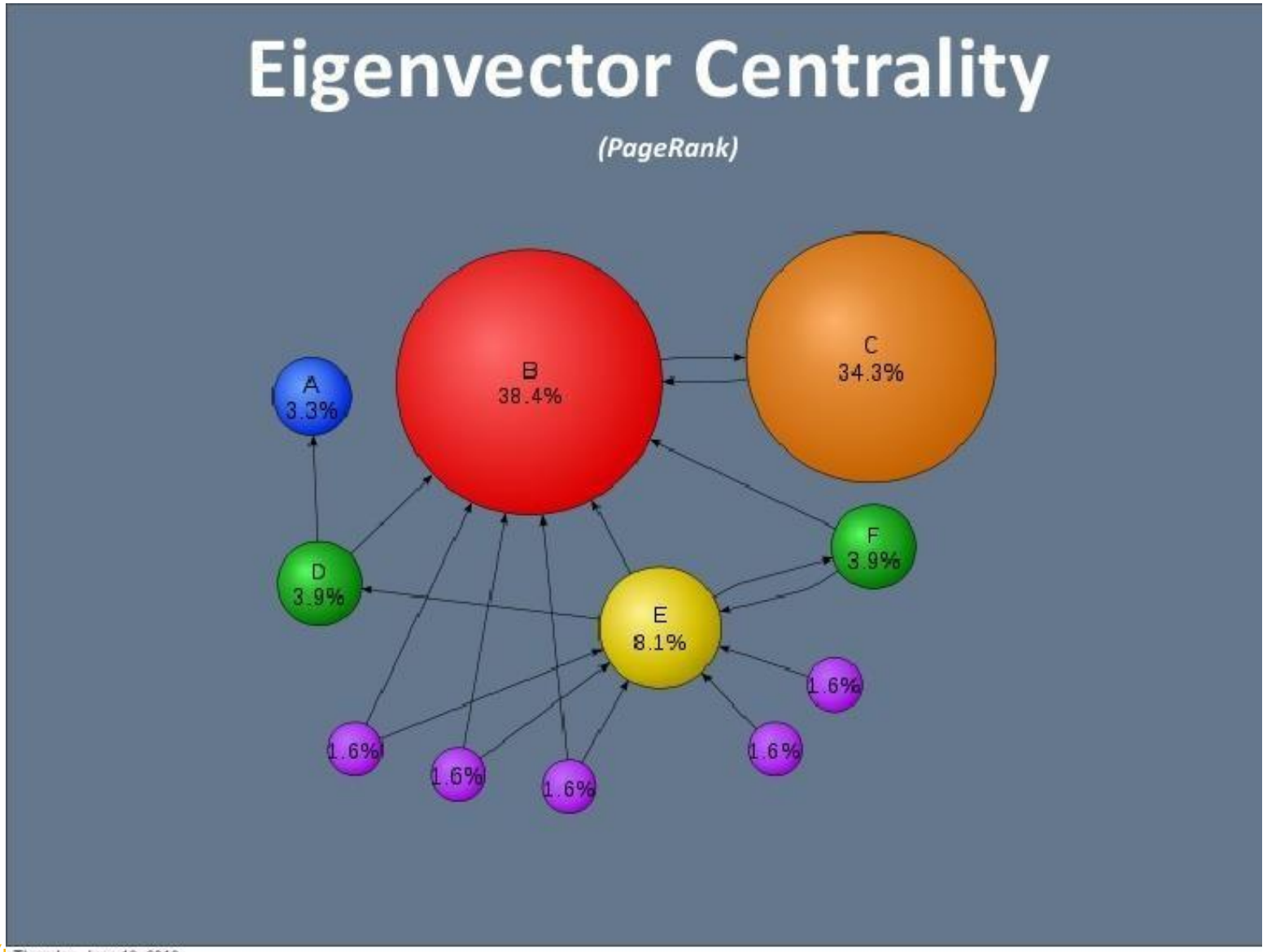
- The edge betweenness centrality refers to the fraction of all pairs of shortest paths of the network that pass through a given edge.
- Computation is more-or-less similar to that of betweenness centrality
- The flow betweenness centrality the fraction of all paths (not necessarily the shortest paths) of the network that passes through a given edge.
- Clearly, flow betweenness centrality measure is computationally expensive than betweenness or edge betweenness centrality measures.

Eigenvector-Based Centrality Measures

* Our previous discussion on different measures of the “centrality "of a node in a network
  - Degree centrality, closeness centrality, betweenness centrality.

* Another, very important class of centrality measures are based on the idea that a node is important if it is connected to other important nodes

* We discuss such eigenvector-based centrality measures, along with an important applications
  - How Google ranks webpages (PageRank)



https://www.slideshare.net/ recommendations-in-rails/63-Eigenvector_Centrality_PageRankThursday_June_10

Eigenvector Centrality

- The eigen vector centrality x_v of a node v in a network $G(V, E)$ is given by

$$x_v = \frac{1}{\lambda} \sum_{t \in N(v)} x_t = \frac{1}{\lambda} \sum_{t \in V} (a_{vt} \times x_t)$$

where λ is the largest eigen value of the matrix $A = (a_{ij})$, the adjacency matrix of the network G

- The largest eigen value λ is obtained by solving the equation

$$A.X = \lambda.X$$

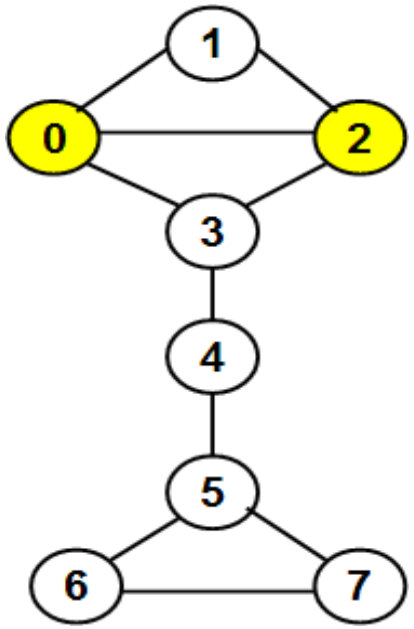
- X above is a column vector, whose v^{th} entry is x_v , the eigen vector centrality of the node v

Eigenvalue and Eigenvector

- Let A be an n x n matrix.
- A scalar λ is called an Eigenvalue of A if there is a nonzero vector X such that **AX = λX.**
- **Therefore (A – λI)X = 0;**
- Such a vector X is called an **Eigenvector of A corresponding to λ**.
- An n x n square matrix has 'n' eigenvalues and the corresponding Eigenvector
- For example, $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is an eigenvector of For matrix $A = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}$ for λ = 4

Eigenvector Centrality: Power Iteration Method:

Example 1



Iteration 1

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 3 \\ 3 \\ 2 \\ 3 \\ 2 \\ 2 \end{bmatrix} \equiv \begin{bmatrix} 0.416 \\ 0.277 \\ 0.416 \\ 0.416 \\ 0.277 \\ 0.416 \\ 0.277 \\ 0.277 \end{bmatrix}$$

Normalized Value = 7.21

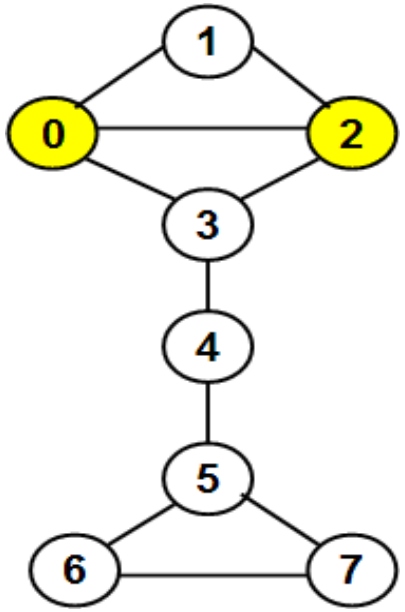
Iteration 2

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0.416 \\ 0.277 \\ 0.416 \\ 0.416 \\ 0.277 \\ 0.416 \\ 0.277 \\ 0.277 \end{bmatrix} = \begin{bmatrix} 1.109 \\ 0.832 \\ 1.109 \\ 1.109 \\ 0.832 \\ 0.831 \\ 0.693 \\ 0.693 \end{bmatrix} \equiv \begin{bmatrix} 0.428 \\ 0.321 \\ 0.428 \\ 0.428 \\ 0.321 \\ 0.321 \\ 0.268 \\ 0.268 \end{bmatrix}$$

Normalized Value = 2.59

Eigenvector Centrality: Power Iteration Method:

Example 1



Iteration 3

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0.428 \\ 0.321 \\ 0.428 \\ 0.428 \\ 0.321 \\ 0.321 \\ 0.268 \\ 0.268 \end{bmatrix} = \begin{bmatrix} 1.177 \\ 0.856 \\ 1.284 \\ 1.177 \\ 0.749 \\ 0.857 \\ 0.589 \\ 0.589 \end{bmatrix} \equiv \begin{bmatrix} 0.441 \\ 0.321 \\ 0.481 \\ 0.441 \\ 0.281 \\ 0.321 \\ 0.221 \\ 0.221 \end{bmatrix}$$

Normalized Value = 2.67

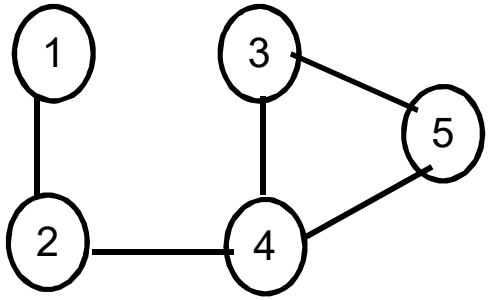
Iteration 4

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0.441 \\ 0.321 \\ 0.481 \\ 0.441 \\ 0.281 \\ 0.321 \\ 0.221 \\ 0.221 \end{bmatrix} = \begin{bmatrix} 1.243 \\ 0.922 \\ 1.203 \\ 1.203 \\ 0.762 \\ 0.723 \\ 0.542 \\ 0.542 \end{bmatrix} \equiv \begin{bmatrix} 0.471 \\ 0.349 \\ 0.456 \\ 0.456 \\ 0.289 \\ 0.274 \\ 0.205 \\ 0.205 \end{bmatrix}$$

Normalized Value = 2.64

Vertex ID	0	1	2	3	4	5	6	7
Principal Eigenvector	0.489	0.364	0.489	0.467	0.264	0.232	0.155	0.155

Eigenvector Centrality: Power Iteration Method: Example 2



$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$\text{Let } X_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Iteration 1

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 3 \\ 2 \end{bmatrix} \equiv \begin{bmatrix} 0.213 \\ 0.426 \\ 0.426 \\ 0.639 \\ 0.426 \end{bmatrix}$$

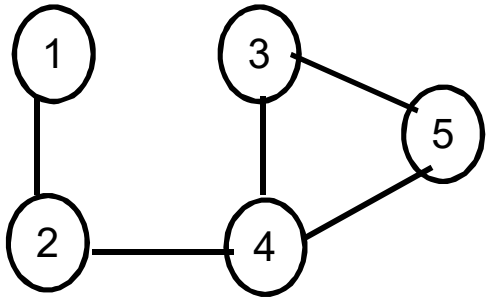
Normalized Value = 4.69

Iteration 2

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.213 \\ 0.426 \\ 0.426 \\ 0.639 \\ 0.426 \end{bmatrix} = \begin{bmatrix} 0.426 \\ 0.852 \\ 1.065 \\ 1.278 \\ 1.065 \end{bmatrix} \equiv \begin{bmatrix} 0.195 \\ 0.389 \\ 0.486 \\ 0.584 \\ 0.486 \end{bmatrix}$$

Normalized Value = 2.19

Eigenvector Centrality: Power Iteration Method: Example 2



$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Let X0 = $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

Iteration 3

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.195 \\ 0.389 \\ 0.486 \\ 0.584 \\ 0.486 \end{bmatrix} = \begin{bmatrix} 0.389 \\ 0.779 \\ 1.07 \\ 1.361 \\ 1.07 \end{bmatrix} \equiv \begin{bmatrix} 0.176 \\ 0.352 \\ 0.484 \\ 0.616 \\ 0.484 \end{bmatrix}$$

Normalized Value = 2.21

Iteration 4

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.176 \\ 0.352 \\ 0.484 \\ 0.616 \\ 0.484 \end{bmatrix} = \begin{bmatrix} 0.352 \\ 0.792 \\ 1.100 \\ 1.320 \\ 1.100 \end{bmatrix}$$

Normalized Value = 2.21 converges

Eigen Vector Centrality

$$\begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} \begin{bmatrix} 0.176 \\ 0.352 \\ 0.484 \\ 0.616 \\ 0.484 \end{bmatrix}$$